Financial Relevance Intelligence Engine

End-to-End Technical Architecture, Multi-Stage Classification Pipeline & System Blueprint

System Purpose: In modern automated financial and contract analysis workflows, feeding every raw document or unvetted text directly into expensive Large Language Models (LLMs) incurs massive compute costs, token waste, and high operational latency. The **Financial Relevance Intelligence Engine (FRIE)** serves as a production-grade filter and extraction preparatory engine. It ingests heterogenous file formats (PDF, DOCX, XLSX, XLS, CSV, TXT, PNG, JPG), determines document and chunk-level financial relevance, pinpoints concrete obligation and transaction passages, and generates structured handoff payloads for downstream financial extraction.

1. High-Level Architectural Pipeline

FRIE processes incoming data through a sequence of six decoupled, idempotent processing stages designed to ensure maximum throughput, robust security, and auditable zero-data-loss execution:

Stage	Name	Core Technology & Strategy
Stage A	Ingestion & Security	SHA-256 deduplication, magic bytes validation, zip-bomb ratio checks, filename sanitization, p
Stage B	Format-Aware Extraction	PyMuPDF (native PDF & layout), RapidOCR/PaddleOCR (scanned pages & images), openpy
Stage C	Structure-Aware Chunking	Sentence boundary chunking with heading breadcrumb preservation, tabular row-slice groupir
Stage D	Reference Dataset & FAISS	Curated 14-category financial ontology + hard negatives. Sentence-Transformers (all-MiniLM-
Stage E	Hybrid Classification	Dual-engine scoring: Inexpensive lexical & obligation pattern heuristics blended with contrastiv
Stage F	Document-Level Aggregation	Ensemble scoring: FINANCIAL_RELEVANT, MIXED_CONTENT, NEEDS_REVIEW, or NOT_

2. Format-Aware Parsing & OCR Strategy

A foundational principle of FRIE is **intelligent format awareness**: never treating structured files as flat text, and never incurring unnecessary OCR overhead on searchable documents.

- **PDF Processing:** Native digital text is extracted via PyMuPDF dict inspection, which retains font metrics, spans, and bounding coordinates. When a page has fewer than 40 native characters, it is recognized as scanned or low-quality and routed to the OCR engine with image deskewing and adaptive contrast enhancements.
- **Spreadsheets (XLSX, XLS, CSV):** Instead of flattening rows into arbitrary text, sheets are split into table regions. The system preserves sheet names, cell coordinates (e.g., Sheet1!B4:F12), headers, formulas (recorded as untrusted text, never evaluated to prevent spreadsheet formula injection), and number formats.
- **Images (PNG, JPG):** Processed through RapidOCR (ONNX Runtime engine) or Tesseract with multi-line coordinate grouping and per-paragraph confidence tracking.
- **Word (DOCX) & Text:** Preserves paragraph indices, heading hierarchies (H1/H2 breadcrumbs), and embedded table cell structures.

3. Financial Reference Dataset & 14-Category Ontology

FRIE relies on a curated, versioned reference dataset (`financial_reference_v1.jsonl`) spanning 14 key financial domains alongside carefully constructed hard-negative non-financial entries:

#	Category Slug	Domain Description
1	income_revenue	Gross/net revenue, subscription earnings, royalties, deferred revenue
2	expenses_operating_costs	OpEx, COGS, SG&A expenses, payroll, operational expenditures
3	accounts_payable_receivable	Vendor ledgers, trade receivables, aging schedules, trade payables
4	invoices_purchase_orders	Purchase order milestones, commercial invoices, line totals, Net 30/60 terms
5	cash_flow_bank_transactions	Operating cash flow, bank statements, wire transfers (SWIFT/ACH), opening balances
6	loans_interest_repayment	Term facilities, promissory notes, principal installments, SOFR/LIBOR margins
7	rent_leases_recurring_payments	Base rent, security deposits, lease escalations, recurring retainers
8	contracts_payment_obligations	Firm covenants: 'shall pay', milestones, consideration, fees for service
9	insurance	Premiums, deductibles, coverage limits, indemnities, claim payouts
10	taxes_duties	Corporate income tax, VAT/GST withholding, statutory duties, customs
11	penalties_late_fees	Default interest, liquidated damages, overdue fees, 1.5% late penalties
12	assets_liabilities_investments	Balance sheet assets, shareholders' equity, goodwill, book/fair value
13	maturity_renewals_deadlines	Debenture maturity dates, auto-renewal covenants, payment deadlines
14	refunds_credits_settlements	Credit memos, warranty settlement payouts, customer chargebacks
15	non_financial (Hard Negatives)	Scientific charge/bond/deposit, $var code, NDAs, boilerplate legal terms

4. The Hybrid Classification & Scoring Engine

Rather than relying strictly on keyword searches or raw semantic embeddings (both of which suffer high false-positive rates when isolated), FRIE implements a **calibrated hybrid engine**:

A. Inexpensive Signal Analysis (Lexical & Grammatical):

Evaluates regex and token patterns without embedding overhead: currency quantities ($4.2M, EUR 12,850.00, Rs. 45,00,000), payment phrases ('shall pay', 'equal monthly installments', 'net 30 days'), financial parties ('Borrower', 'Lessor', 'Tenant'), rate indicators ('SOFR + 2.75%', '8.5% p.a.'), and table headers. Crucially, it applies negative suppression for code syntax (e.g. `$HOME`, `${var}`, regex `^$`) and non-financial polysemy (e.g. 'electric charge', 'chemical bond', 'mineral deposit').

B. Semantic Embedding & Category Similarity:

Generates L2-normalized 384-dimensional dense vectors using `sentence-transformers/all-MiniLM-L6-v2`. Computes cosine similarity across the FAISS vector index. The score is calculated as a **contrastive multi-example aggregation** (mean of top-2 matches in a category minus the top hard-negative matches), preventing single matching words from polluting relevance.

C. Calibrated Three-Way Decision:

Each chunk receives an authoritative decision based on empirically validated thresholds: **RELEVANT (Score ≥ 0.60)**, **UNCERTAIN (0.35 < Score < 0.60)**, or **IRRELEVANT (Score ≤ 0.35)**. Uncertain chunks can optionally invoke an isolated secondary LLM tie-breaker with strict JSON output schemas.

5. Document-Level Aggregation & Downstream Handoff

Chunk-level verdicts are assembled into an overall document state:

- **FINANCIAL_RELEVANT:** Dominantly financial documents (e.g. quarterly balance sheets, loan agreements).
- **MIXED_CONTENT:** Long commercial contracts or company announcements that contain both non-financial sections (e.g., NDA clauses, governing law) and concrete financial commitments. *Only relevant regions are prepared for downstream LLM extraction*, preventing massive token waste.
- **NEEDS_REVIEW:** Flagged when all chunks appear non-financial but OCR confidence is substandard or pages have sparse text, safeguarding against false negatives.
- **NOT_FINANCIAL:** Cleanly filtered documents preserved in the audit log without LLM invocation.

6. Operational Interface & User Experience

The system provides a responsive, production-ready React (TypeScript) dashboard running on Vite:

Module	Route	Capabilities & Features
Dashboard	/	Telemetry cards, document status breakdown, average latency, embedding cache hit-rates, chunk c
Upload	/upload	Multi-file drag-and-drop & native file picker, client-side format & size validation (50MB cap), real-time
Document Results	/documents/:id	Document status timeline, overall relevance badge, heuristic confidence breakdown, extraction qual
Chunk Explorer	/documents/:id/chunks	Interactive passage inspector with signal highlighting, source bounding coordinates, table structure r
Reference Dataset	/reference	Vector database browser, category distribution metrics, dynamic entry editor, and one-click vector in
Evaluation	/evaluation	Precision-Recall curves, latency benchmarks, OCR failure metrics, and per-format performance stat

7. Enterprise Security & Container Deployment

- **Prompt Injection Isolation:** Extracted text from uploaded documents is strictly treated as untrusted data. It is never interpolated into executable scripts or unescaped HTML.
- **Local Offline Execution:** Both the embedding models (Sentence-Transformers) and the vector index (FAISS) run entirely locally on the server CPU. No document content is uploaded to third-party APIs.
- **Macro & Archive Hardening:** Rejects macro-enabled Office extensions (`.docm`, `.xlsm`) and evaluates zip-bomb uncompressed size ratios before parsing.
- **Dockerized Architecture:** A multi-container `docker-compose.yml` configures the FastAPI backend on port 8000 and the production Nginx web server on port 5173.

System Status	Live & Hosted Locally
Backend API	http://localhost:8000 (FastAPI / OpenAPI)
Frontend UI	http://localhost:5173 (React / Vite)
Active Model	sentence-transformers/all-MiniLM-L6-v2 | FAISS CPU Index
OCR Engine	RapidOCR (ONNX Runtime)